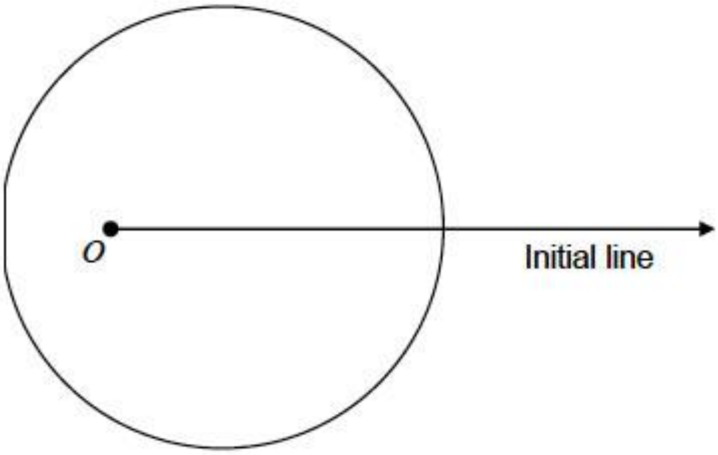
Q1.

The diagram shows a sketch of a curve C, the pole O and the initial line.



The polar equation of C is $r = 4 + 2\cos\theta$, $\quad -\pi \leq \theta \leq \pi$

(a) Show that the area of the region bounded by the curve C is 18π

(4)

(b) Points A and B lie on the curve C such that $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$ and AOB is an equilateral triangle.

Find the polar equation of the line segment AB

(4)

(Total 8 marks)

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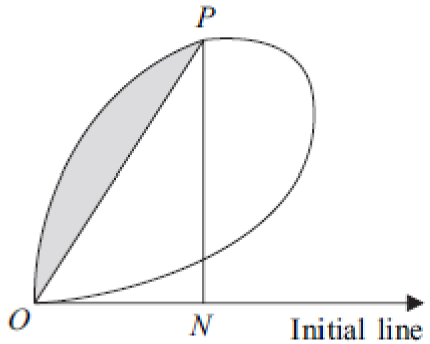
Q2.

The diagram shows a sketch of a curve.



The polar equation of the curve is

$$r = \sin 2\theta\sqrt{\left(2 + \frac{1}{2}\cos\theta\right)}, \quad 0 \leqslant \theta \leqslant \frac{\pi}{2}$$

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The point P is the point of the curve at which $\theta = \frac{\pi}{3}$.

The perpendicular from P to the initial line meets the initial line at the point N.

(a) (i) Find the exact value of r when $\theta = \frac{\pi}{3}$.

(2)

(ii) Show that the polar equation of the line PN is $r = \frac{3\sqrt{3}}{8} \sec \theta$.

(2)

(iii) Find the area of triangle ONP in the form $\frac{k\sqrt{3}}{128}$, where k is an integer.

(2)

(b) (i) Using the substitution $u = \sin \theta$, or otherwise, find $\int \sin^n \theta \cos \theta \, d\theta$, where $n \geq 2$.

(2)

(ii) Find the area of the shaded region bounded by the line OP and the arc OP of the curve. Give your answer in the form $a\pi + b\sqrt{3} + c$, where a, b and c are constants.

(8)

(Total 16 marks)

Q3.

The diagram shows a sketch of a curve and a circle.

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The polar equation of the curve is

$$r = 3 + 2 \sin \theta, \quad 0 \leq \theta \leq 2\pi$$

The circle, whose polar equation is $r = 2$, intersects the curve at the points P and Q, as shown in the diagram.

(a) Find the polar coordinates of P and the polar coordinates of Q.

(3)

(b) A straight line, drawn from the point P through the pole O, intersects the curve

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again at the point A.

(i) Find the polar coordinates of A.

(2)

(ii) Find, in surd form, the length of AQ.

(3)

(iii) Hence, or otherwise, explain why the line AQ is a tangent to the circle $r = 2$.

(2)

(c) Find the area of the shaded region which lies inside the circle $r = 2$ but outside the curve
$r = 3 + 2 \sin \theta$. Give your answer in the form $\frac{1}{6}\left(m\sqrt{3} + n\pi\right)$, where m and n are integers.

(9)

(Total 19 marks)

Q4.

The diagram shows a sketch of the curve C with polar equation

$$r = 3 + 2 \cos \theta, \qquad 0 \leq \theta \leq 2\pi$$

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(a) Find the area of the region bounded by the curve C.

(6)

(b) A circle, whose cartesian equation is $(x - 4)^2 + y^2 = 16$, intersects the curve C at the points A and B.

(i) Find, in surd form, the length of AB.

(6)

(i) Find the perimeter of the segment AOB of the circle, where O is the pole.

(3)

(Total 12 marks)

Q5.

The diagram shows a sketch of a curve C, the pole O and the initial line.

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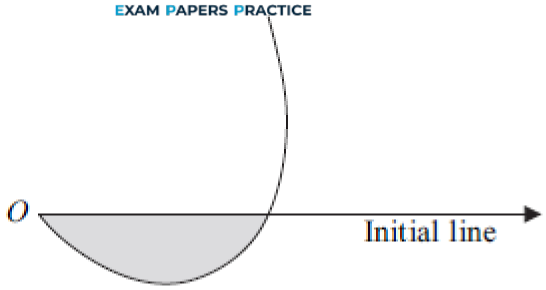


The polar equation of C is

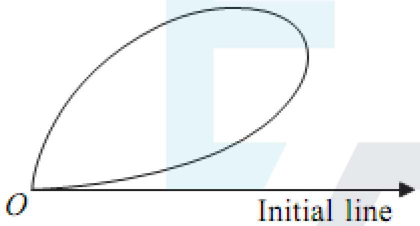
$$r = 2\sqrt{1 + \tan\theta}, \; -\frac{\pi}{4} \leqslant \theta \leqslant \frac{\pi}{4}$$

Show that the area of the shaded region, bounded by the curve C and the initial line, is $\frac{\pi}{2}$ – ln 2.

(Total 4 marks)

Q6.

The diagram shows a sketch of a curve C.



The polar equation of the curve is

$$r = 2 \sin 2\theta\sqrt{\cos\theta}, \quad 0 \leq \theta \leq \frac{\pi}{2}$$

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Show that the area of the region bounded by C is $\frac{16}{15}$.

(Total 7 marks)

Q7.

The curve C_1 is defined by $r = 2 \sin \theta$, $\quad 0 \leq \theta < \frac{\pi}{2}$.

The curve C_2 is defined by $r = \tan \theta$, $\quad 0 \leq \theta < \frac{\pi}{2}$.

(a) Find a cartesian equation of C_1

(3)

(b) (i) Prove that the curves C_1 and C_2 meet at the pole O and at one other point, P, in the given domain. State the polar coordinates of P.

(4)

(ii) The point A is the point on the curve C_1 at which $\theta = \frac{\pi}{4}$.

The point B is the point on the curve C_2 at which $\theta = \frac{\pi}{4}$.

Determine which of the points A or B is further away from the pole O, justifying your answer.

(2)

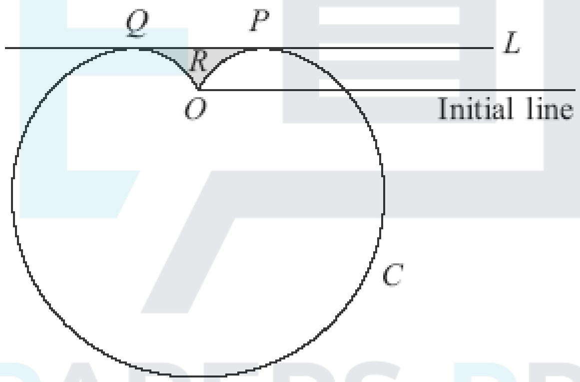
(iii) Show that the area of the region bounded by the arc OP of C_1 and the arc OP of C_2 is $a\pi + b\sqrt{3}$, where a and b are rational numbers.

(10)

(Total 19 marks)

Q8.

The diagram shows a sketch of a curve C and a line L, which is parallel to the initial line and touches the curve at the points P and Q.



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The polar equation of the curve C is

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$$r = 4(1 - \sin\theta), \qquad 0 \leq \theta < 2\pi$$

and the polar equation of the line L is

$$r \sin\theta = 1$$

(a) Show that the polar coordinates of P are $\left(2, \frac{\pi}{6}\right)$ and find the polar coordinates of Q.

(5)

(b) Find the area of the shaded region R bounded by the line L and the curve C. Give your answer in the form $m\sqrt{3} + n\pi$, where m and n are integers.

(11)

(Total 16 marks)

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Q9.

The polar equation of a curve C_1 is

$$r = 2(\cos\theta - \sin\theta),\ 0 \le \theta \le 2\pi$$

(a) (i) Find the cartesian equation of C_1.

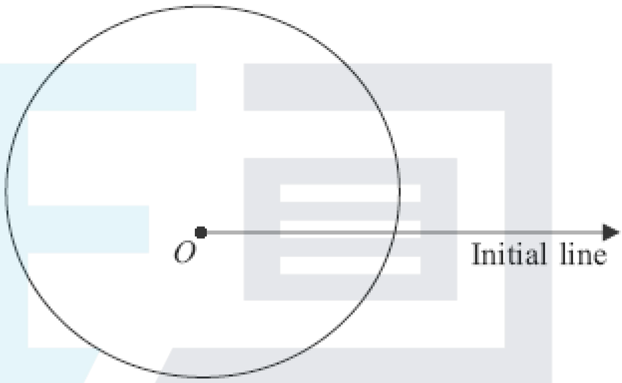
(4)

(ii) Deduce that C_1 is a circle and find its radius and the cartesian coordinates of its centre.

(3)

(b) The diagram shows the curve C_2 with polar equation

$$r = 4 + \sin\theta,\quad 0 \le \theta \le 2\pi$$



(i) Find the area of the region that is bounded by C_2.

(6)

(ii) Prove that the curves C_1 and C_2 do not intersect.

(4)

(iii) Find the area of the region that is outside C_1 but inside C_2.

(2)

(Total 19 marks)

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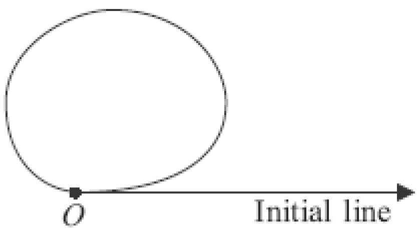
Q10.

The diagram shows a sketch of a loop, the pole O and the initial line.



The polar equation of the loop is

$$r = (2 + \cos\theta)\sqrt{\sin\theta},\ 0 \le \theta \le \pi$$

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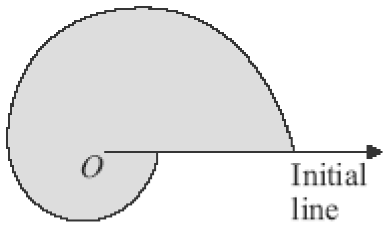
Find the area enclosed by the loop.

(Total 6 marks)

Q11.

The diagram shows the curve C_1 with polar equation

$$r = 1 + 6e^{-\frac{\theta}{\pi}}, \quad 0 \leq \theta \leq 2\pi$$



(a) Find, in terms of π and e, the area of the shaded region bounded by C_1 and the initial line.

(5)

(b) The polar equation of a curve C_2 is

$$r = e^{\frac{\theta}{\pi}}, \quad 0 \leq \theta \leq 2\pi$$

Sketch the curve C_2 and state the polar coordinates of the end-points of this curve.

(4)

(c) The curves C_1 and C_2 intersect at the point P. Find the polar coordinates of P.

(5)

(Total 14 marks)

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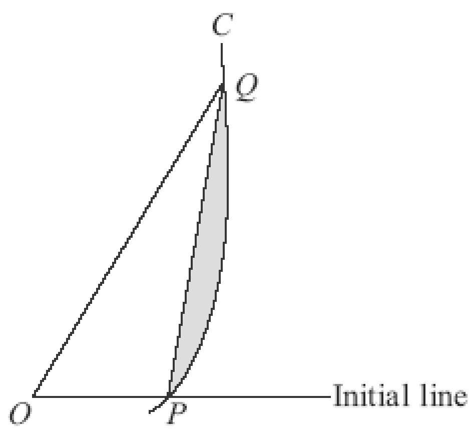
Q12.

The diagram shows a sketch of part of the curve C whose polar equation is $r = 1 + \tan\theta$. The point O is the pole.



The points P and Q on the curve are given by $\theta = 0$ and $\theta = \frac{\pi}{3}$ respectively.

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(a) Show that the area of the region bounded by the curve C and the lines OP and OQ is

$$\frac{1}{2}\sqrt{3} + \ln 2$$

(6)

(b) Hence find the area of the shaded region bounded by the line PQ and the arc PQ of C.

(3)

(Total 9 marks)

Q13.

The polar equation of a curve C is

$$r = 5 + 2\cos\theta, \qquad -\pi \leq \theta \leq \pi$$

(a) Verify that the points A and B, with **polar coordinates** $(7, 0)$ and $(3, \pi)$ respectively, lie on the curve C.

(2)

(b) Sketch the curve C.

(2)

(c) Find the area of the region bounded by the curve C.

(6)

(d) The point P is the point on the curve C for which $\theta = \alpha$, where $0 < \alpha \leq \frac{\pi}{2}$. The point Q lies on the curve such that POQ is a straight line, where the point O is the pole. Find, in terms of α, the area of triangle OQB.

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(4)

(Total 14 marks)

Q14.

A curve C has polar equation

$r = 6 + 4\cos\theta \qquad -\pi \leq \theta \leq \pi$

The diagram shows a sketch of the curve C, the pole O and the initial line.

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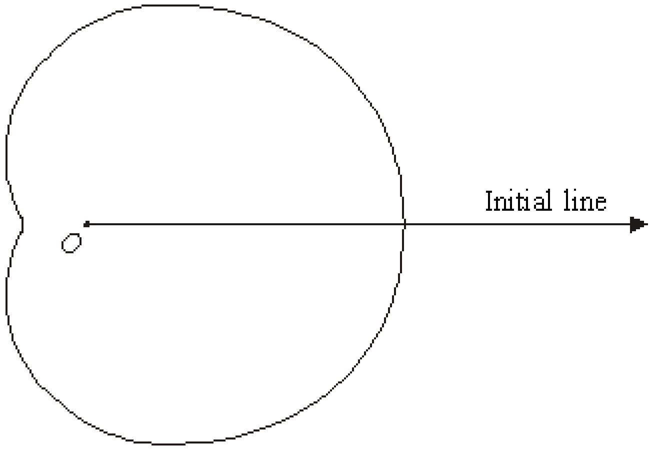

(a) Calculate the area of the region bounded by the curve C.

(6)

(b) The point P is the point on the curve C for which $\theta = \frac{2\pi}{3}$.

The point Q is the point on C for which $\theta = \pi$.

Show that QP is parallel to the line $\theta = \frac{\pi}{2}$.

(4)

(c) The line PQ intersects the curve C again at a point R.

The line RO intersects C again at a point S

(i) Find, in surd form, the length of PS.

(4)

(ii) Show that the angle OPS is a right angle.

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(1)

(Total 15 marks)

Q15.

(a) Show that $(\cos\theta + \sin\theta)^2 = 1 + \sin 2\theta$.

(1)

(b) A curve has cartesian equation

$(x^2 + y^2)^3 = (x + y)^4$

Given that $r \geq 0$, show that the polar equation of the curve is

$r = 1 + \sin 2\theta$

(4)

(c) The curve with polar equation

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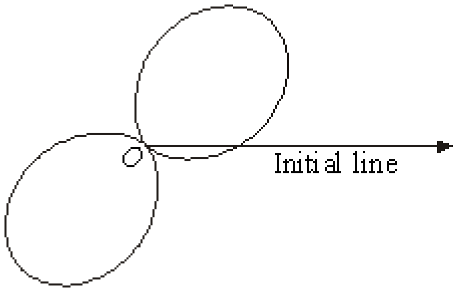
$r = 1 + \sin 2\theta, \qquad -\pi \leq \theta \leq \pi$

is shown in the diagram.



(i) Find the two values of θ for which $r = 0$.

(3)

(ii) Find the area of one of the loops.

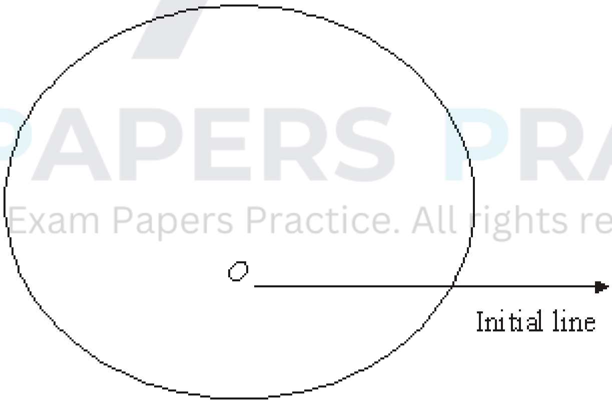
(6)
(Total 14 marks)

Q16.

(a) A circle C_1 has cartesian equation $x^2 + (y - 6)^2 = 36$. Show that the polar equation of C_1 is $r = 12 \sin \theta$.

(4)

(b) A curve C_2 with polar equation $r = 2\sin \theta + 5$, $0 \leq \theta \leq 2\pi$ is shown in the diagram.



Calculate the area bounded by C_2.

(6)

(c) The circle C_1 intersects the curve C_2 at the points P and Q. Find, in surd form, the area of the quadrilateral $OPMQ$, where M is the centre of the circle and O is the pole.

(6)
(Total 16 marks)

Q17.

The diagram shows the curve C with polar equation

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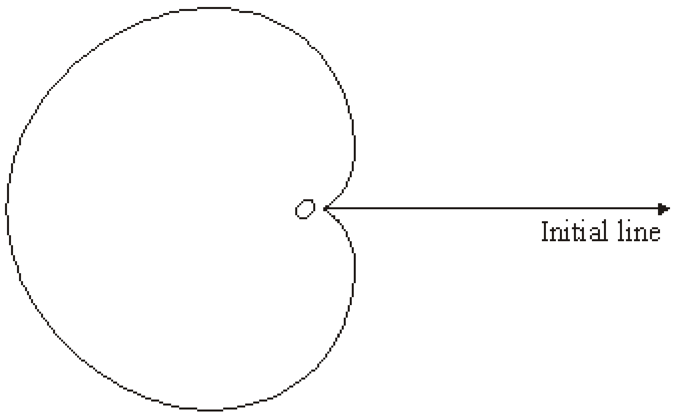
$r = 6(1 - \cos \theta)$, $0 \le \theta < \pi$



(a) Find the area of the region bounded by the curve C.

(6)

(b) The circle with cartesian equation $x^2 + y^2 = 9$ intersects the curve C at the points A and B.

(i) Find the polar coordinates of A and B.

(4)

(ii) Find, in surd form, the length of AB.

(2)

(Total 12 marks)

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